INNOMATICS®
RESEARCH LABS

INNOVATION. AUTOMATION. ANALYTICS

PROJECT ON

MUSIC STORE DATA ANALYSIS

About US

- **Group member 1: K. Dileep, B. Tech in CSE**
  ID: , Batch: 448

- **Group member 2: K. Nikshith, B. Tech in CSE**
  ID: , Batch: 448

- **Group member 3: M. Lakshmikar, MBA**
  ID: , Batch: 448

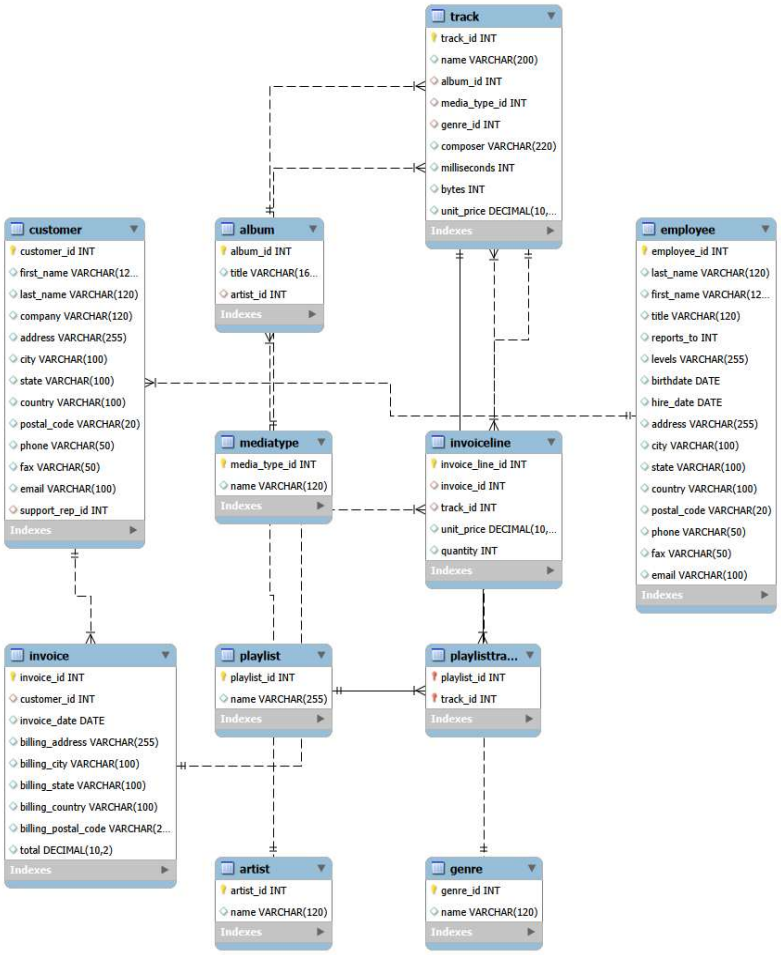
INNOMATICS RESEARCH LABS

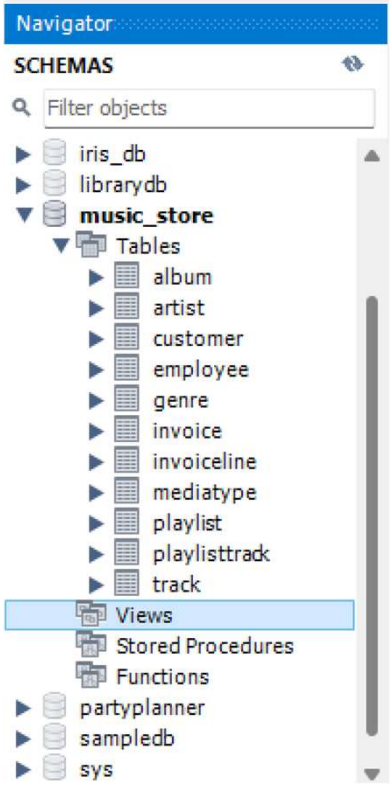
Objective of the Project

- To analyze a Music Store database using SQL
- To understand and apply relational database concepts
- To extract business insights from raw data
- To practice joins, aggregations, and subqueries
- To improve real-world SQL data analysis skills

INNOMATICS RESEARCH LABS

ER Diagram and Schema Explanation





INNOMATICS RESEARCH LABS

ER Diagram and Schema Explanation

The Music Store Database follows a Relational Schema Design

<u>Key Tables</u>

- Employee – Stores employee details and hierarchy
- Customer – Stores customer information
- Invoice – Records purchase transactions
- InvoiceLine – Stores individual track purchases
- Track – Contains song details
- Album – Groups tracks under albums
- Artist – Stores artist information
- Genre – Classifies music genres

INNOMATICS RESEARCH LABS

ER Diagram and schema explanation

The Music Store Database follows a Relational Schema Design

Relationships

- One customer can have multiple invoices
- One invoice can have multiple invoice lines
- Each track belongs to one genre and one album
- Each album is linked to one artist

INNOMATICS RESEARCH LABS

Key analysis questions (use cases)

The questions were centered around

1. Employee Hierarchy Analysis.
2. Geographical Invoice Distribution.
3. High-Value Transactions.
4. Revenue by City.
5. Best Customer Identification.
6. Genre-Based Customer Preferences (Rock Listeners).
7. Top Rock Artists by Track Count.
8. Track Duration Analysis.
9. Customer Spending by Artist.
10. Most Popular Genre by Country.
11. Top-Spending Customer by Country.

INNOMATICS RESEARCH LABS

SQL query results with screenshots or summaries

1. Who is the senior most employee based on job title?

```sql
CREATE TABLE JobTitleRank (
  title VARCHAR(120) PRIMARY KEY,
  rank_order INT NOT NULL
);

INSERT INTO JobTitleRank (title, rank_order) VALUES
('Intern', 1),
('Junior Associate', 2),
('IT Staff', 3),
('Sales Support Agent', 4),
('Senior Support Agent', 5),
('Team Lead', 6),
('IT Manager', 7),
('Sales Manager', 8),
('Senior Manager', 9),
('Assistant General Manager', 10),
('General Manager', 11),
('Senior General Manager', 12),
('Director', 13),
('Vice President', 14),
('CEO', 15);
```

```sql
SELECT CONCAT(e.first_name, ' ', e.last_name) AS full_name
FROM Employee e
JOIN JobTitleRank j
  ON e.title = j.title
ORDER BY j.rank_order DESC
LIMIT 1;
```

	full_name
▶	Mohan Madan

INNOMATICS RESEARCH LABS

2. Which countries have the most Invoices?

```sql
SELECT
    billing_country,
    COUNT(*) AS total_invoices
FROM invoice
GROUP BY billing_country
ORDER BY total_invoices DESC
LIMIT 5;
```

billing_country	total_invoices
USA	131
Canada	76
Brazil	61
France	50
Germany	41

INNOMATICS RESEARCH LABS

3. What are the top 3 values of total invoice?

```
SELECT total
FROM invoice
ORDER BY total DESC
LIMIT 3;
```

	total
▶	23.76
	19.80
	19.80

INNOMATICS
RESEARCH LABS

4. Which city has the best customers? - We would like to throw a promotional Music Festival in the city we made the most money. Write a query that returns one city that has the highest sum of invoice totals. Return both the city name & sum of all invoice totals.

```
SELECT billing_city,
       SUM(total) AS total_revenue
FROM invoice
GROUP BY billing_city
ORDER BY total_revenue DESC
LIMIT 1;
```

	billing_city	total_revenue
▶	Prague	273.24

INNOMATICS RESEARCH LABS

5. Who is the best customer? - The customer who has spent the most money will be declared the best customer. Write a query that returns the person who has spent the most money.

```
SELECT c.first_name,
       c.last_name,
       SUM(i.total) AS total_spent
FROM customer c
JOIN invoice i
  ON c.customer_id = i.customer_id
GROUP BY c.customer_id
ORDER BY total_spent DESC
LIMIT 1;
```

	first_name	last_name	total_spent
▶	František	Wichterlová	144.54

INNOMATICS RESEARCH LABS

6. Write a query to return the email, first name, last name, & Genre of all Rock Music listeners. Return your list ordered alphabetically by email starting with A.

```
SELECT DISTINCT
       c.email,
       c.first_name,
       c.last_name,
       g.name AS genre
FROM customer c
JOIN invoice i  ON c.customer_id = i.customer_id
JOIN invoiceline il ON i.invoice_id = il.invoice_id
JOIN track t ON il.track_id = t.track_id
JOIN genre g ON t.genre_id = g.genre_id
WHERE LOWER(TRIM(g.name)) LIKE '%rock%'
ORDER BY c.email;
```

email	first_name	last_name	genre
aaronmitchell@yahoo.ca	Aaron	Mitchell	Rock
alero@uol.com.br	Alexandre	Rocha	Rock
astrid.gruber@apple.at	Astrid	Gruber	Rock
bjorn.hansen@yahoo.no	Bjørn	Hansen	Rock
camille.bernard@yahoo.fr	Camille	Bernard	Rock
daan_peeters@apple.be	Daan	Peeters	Rock
diego.gutierrez@yahoo.ar	Diego	Gutiérrez	Rock
dmiller@comcast.com	Dan	Miller	Rock
dominiquelefebvre@gmail.com	Dominique	Lefebvre	Rock
edfrancis@yachoo.ca	Edward	Francis	Rock
eduardo@woodstock.com.br	Eduardo	Martins	Rock
ellie.sullivan@shaw.ca	Ellie	Sullivan	Rock
emma_jones@hotmail.com	Emma	Jones	Rock
enrique_munoz@yahoo.es	Enrique	Muñoz	Rock
fernadaramos4@uol.com.br	Fernanda	Ramos	Rock
fharris@google.com	Frank	Harris	Rock
fralston@gmail.com	Frank	Ralston	Rock
frantisekw@jetbrains.com	František	Wichterlová	Rock
ftremblay@gmail.com	François	Tremblay	Rock
fzimmermann@yahoo.de	Fynn	Zimmerm...	Rock
hannah.schneider@yahoo.de	Hannah	Schneider	Rock
hholy@gmail.com	Helena	Holý	Rock
hleacock@gmail.com	Heather	Leacock	Rock
hughoreilly@apple.ie	Hugh	O'Reilly	Rock

INNOMATICS RESEARCH LABS

7. Let's invite the artists who have written the most rock music in our dataset. Write a query that returns the Artist name and total track count of the top 10 rock bands.

```
SELECT
       ar.name AS artist_name,
       COUNT(t.track_id) AS total_rock_tracks
FROM artist ar
JOIN album al
  ON ar.artist_id = al.artist_id
JOIN track t
  ON al.album_id = t.album_id
JOIN genre g
  ON t.genre_id = g.genre_id
WHERE LOWER(TRIM(g.name)) LIKE '%rock%'
GROUP BY ar.artist_id, ar.name
ORDER BY total_rock_tracks DESC
LIMIT 10;
```

artist_name	total_rock_tracks
Led Zeppelin	114
U2	112
Deep Purple	92
Iron Maiden	81
Pearl Jam	54
Van Halen	52
Queen	45
The Rolling Stones	41
Creedence Clearwater Revival	40
Kiss	35

INNOMATICS RESEARCH LABS

8. Return all the track names that have a song length longer than the average song length.- Return the Name and Milliseconds for each track. Order by the song length, with the longest songs listed first.

```
SELECT
        name,
        milliseconds
FROM track
WHERE milliseconds > (
    SELECT AVG(milliseconds)
    FROM track
)
ORDER BY milliseconds DESC;
```

name	milliseconds
Occupation / Precipice	5286953
Through a Looking Glass	5088838
Greetings from Earth, Pt. 1	2960293
The Man With Nine Lives	2956998
Battlestar Galactica, Pt. 2	2956081
Battlestar Galactica, Pt. 1	2952702
Murder On the Rising Star	2935894
Battlestar Galactica, Pt. 3	2927802
Take the Celestra	2927677
Fire In Space	2926593
The Long Patrol	2925008
The Magnificent Warriors	2924716
The Living Legend, Pt. 1	2924507
The Gun On Ice Planet Zer...	2924341
The Hand of God	2924007
Experiment In Terra	2923548
War of the Gods, Pt. 2	2923381
The Living Legend, Pt. 2	2923298
War of the Gods, Pt. 1	2922630
Lost Planet of the Gods, P...	2922547
Baltar's Escape	2922088
The Lost Warrior	2920045
Lost Planet of the Gods, P...	2914664

INNOMATICS RESEARCH LABS

9. Find how much amount is spent by each customer on artists? Write a query to return customer name, artist name and total spent

```sql
SELECT
       CONCAT(c.first_name, ' ', c.last_name) AS customer_name,
       ar.name AS artist_name,
       SUM(il.unit_price * il.quantity) AS total_spent
FROM customer c
JOIN invoice i
  ON c.customer_id = i.customer_id
JOIN invoiceline il
  ON i.invoice_id = il.invoice_id
JOIN track t
  ON il.track_id = t.track_id
JOIN album al
  ON t.album_id = al.album_id
JOIN artist ar
  ON al.artist_id = ar.artist_id
GROUP BY c.customer_id, ar.artist_id
ORDER BY customer_name, total_spent DESC;
```

customer_name	artist_name	total_spent
Aaron Mitchell	James Brown	19.80
Aaron Mitchell	Chris Cornell	13.86
Aaron Mitchell	Men At Work	1.98
Aaron Mitchell	Nirvana	1.98
Aaron Mitchell	Pearl Jam	1.98
Aaron Mitchell	Creedence Clearwater Revival	1.98
Aaron Mitchell	U2	1.98
Aaron Mitchell	Guns N' Roses	0.99
Aaron Mitchell	Alanis Morissette	0.99
Aaron Mitchell	Soundgarden	0.99
Aaron Mitchell	JET	0.99
Aaron Mitchell	Audioslave	0.99
Aaron Mitchell	Miles Davis	0.99
Aaron Mitchell	Jota Quest	0.99
Aaron Mitchell	Red Hot Chili Peppers	0.99
Aaron Mitchell	Amy Winehouse	0.99
Aaron Mitchell	Godsmack	0.99
Aaron Mitchell	Green Day	0.99
Aaron Mitchell	Rush	0.99
Aaron Mitchell	AC/DC	0.99
Aaron Mitchell	Black Sabbath	0.99
Aaron Mitchell	The Rolling Stones	0.99
Aaron Mitchell	Aerosmith	0.99
Aaron Mitchell	Pink Floyd	0.99

INNOMATICS RESEARCH LABS

10. We want to find out the most popular music Genre for each country. We determine the most -- popular genre as the genre with the highest number of purchases. Write a query that returns each country along with the top Genre. For countries where the maximum number of purchases is shared, return all Genres

```sql
SELECT
    i.billing_country AS country,
    g.name AS genre,
    COUNT(il.invoice_line_id) AS purchase_count
FROM invoice i
JOIN invoiceline il
  ON i.invoice_id = il.invoice_id
JOIN track t
  ON il.track_id = t.track_id
JOIN genre g
  ON t.genre_id = g.genre_id
GROUP BY i.billing_country, g.genre_id, g.name;
```

```sql
WITH genre_purchases AS (
    SELECT
        i.billing_country AS country,
        g.name AS genre,
        COUNT(il.invoice_line_id) AS purchase_count,
        DENSE_RANK() OVER (
            PARTITION BY i.billing_country
            ORDER BY COUNT(il.invoice_line_id) DESC
        ) AS genre_rank
    FROM invoice i
    JOIN invoiceline il
      ON i.invoice_id = il.invoice_id
    JOIN track t
      ON il.track_id = t.track_id
    JOIN genre g
      ON t.genre_id = g.genre_id
    GROUP BY i.billing_country, g.genre_id, g.name
)
SELECT
    country,
    genre,
    purchase_count
FROM genre_purchases
WHERE genre_rank = 1
ORDER BY country;
```

country	genre	purchase_count
Argentina	Alternative & Punk	17
Australia	Rock	34
Austria	Rock	40
Belgium	Rock	26
Brazil	Rock	205
Canada	Rock	333
Chile	Rock	61
Czech Republic	Rock	143
Denmark	Rock	24
Finland	Rock	46
France	Rock	211
Germany	Rock	194
Hungary	Rock	44
India	Rock	102
Ireland	Rock	72
Italy	Rock	35
Netherlands	Rock	33
Norway	Rock	40
Poland	Rock	40
Portugal	Rock	108
Spain	Rock	46
Sweden	Rock	60
United Kingdom	Rock	166
USA	Rock	561

INNOMATICS RESEARCH LABS

11. Write a query that determines the customer that has spent the most on music for each country. Write a query that returns the country along with the top customer and how much they spent. For countries where the top amount spent is shared, provide all customers who spent this amount

```sql
WITH customer_spending AS (
    SELECT
        i.billing_country AS country,
        c.customer_id,
        CONCAT(c.first_name, ' ', c.last_name) AS customer_name
        SUM(il.unit_price * il.quantity) AS total_spent,
        DENSE_RANK() OVER (
            PARTITION BY i.billing_country
            ORDER BY SUM(il.unit_price * il.quantity) DESC
        ) AS spending_rank
    FROM customer c
    JOIN invoice i
      ON c.customer_id = i.customer_id
    JOIN invoiceline il
      ON i.invoice_id = il.invoice_id
    GROUP BY
        i.billing_country,
        c.customer_id,
        c.first_name,
        c.last_name
)
```

```sql
SELECT
    country,
    customer_name,
    total_spent
FROM customer_spending
WHERE spending_rank = 1
ORDER BY country;
```

country	customer_name	total_spent
Argentina	Diego Gutiérrez	39.60
Australia	Mark Taylor	81.18
Austria	Astrid Gruber	69.30
Belgium	Daan Peeters	60.39
Brazil	Luís Gonçalves	108.90
Canada	François Tremblay	99.99
Chile	Luis Rojas	97.02
Czech Republic	František Wichterlová	144.54
Denmark	Kara Nielsen	37.62
Finland	Terhi Hämäläinen	79.20
France	Wyatt Girard	99.99
Germany	Fynn Zimmermann	94.05
Hungary	Ladislav Kovács	78.21
India	Manoj Pareek	111.87
Ireland	Hugh O'Reilly	114.84
Italy	Lucas Mancini	50.49
Netherlands	Johannes Van der Berg	65.34
Norway	Bjørn Hansen	72.27
Poland	Stanisław Wójcik	76.23
Portugal	João Fernandes	102.96
Spain	Enrique Muñoz	98.01
Sweden	Joakim Johansson	75.24
United Kingdom	Phil Hughes	98.01
USA	Jack Smith	98.01

INNOMATICS RESEARCH LABS

Final business insights and recommendations

Key Business Insights

- The USA generates the highest overall revenue among all countries
- India shows growing customer engagement, indicating strong potential for market expansion
- Rock music emerges as the most popular genre across multiple countries
- A small group of customers contributes a significant portion of total revenue
- Certain cities consistently record high invoice totals, reflecting strong regional demand

INNOMATICS RESEARCH LABS

Final business insights and recommendations

Business Recommendations

- Organize promotional music events in high-revenue cities across the USA and India
- Launch region-specific marketing strategies tailored to Indian customer preferences
- Focus targeted campaigns on Rock genre listeners
- Introduce loyalty and reward programs for high-spending customers

INNOMATICS RESEARCH LABS

Conclusion

- This project helped us understand how SQL can be used to analyze real-world business data.
- By applying JOINS, AGGREGATIONS, and SUBQUERIES, we were able to extract meaningful business insights.
- The analysis demonstrates how data-driven decision-making can significantly improve business strategy and performance.

INNOMATICS RESEARCH LABS

Our Experience/Challenges working on SQL – Data Analysis Project.

- Understanding complex table relationships and foreign key dependencies was challenging in the initial stages
- While importing CSV data, we faced issues related to data type mismatches, NULL values, and date format errors
- Writing optimized JOIN queries across multiple tables required careful planning and repeated testing
- Handling aggregate functions, GROUP BY, and nested subqueries improved our logical thinking and SQL proficiency
- Debugging incorrect query outputs and improving query performance strengthened our real-world data analysis confidence

INNOMATICS RESEARCH LABS

Q&A Slide

- We are happy to ANSWER your questions

INNOMATICS RESEARCH LABS

THANK YOU

INNOMATICS
RESEARCH LABS